

CS111 Assign1 Write-Up

Name: Ronil Prasad

Student Number: S11231541

Exercise 1

(a) Pseudocode

1. Initialization:

- Define constants for current year, lowest birth year, income brackets, tax rates, and residential flat rates.
- Declare variables for user information (first name, last name, gender, initial, employee ID, birth year, age, residence status), income, tax, and user input for recalculating.

2. Welcome Message:

- Print a welcome message for the Tax Calculator.

3. Do-While Loop:

3.1 User Information:

- Prompt the user for their first name, read it, and extract the initial.
- Prompt the user for their last name and read it.
- Loop until a valid gender (M, F, or O) is entered:
 - Prompt the user for their gender.
 - Read the gender and convert it to uppercase.
 - Validate gender input.
- Prompt the user for their Employee ID and read it.
- Loop until a valid birth year is entered:
 - Prompt the user for their birth year.
 - Read the birth year.
 - If the input is invalid (not an integer) or outside the allowed range, display an error message and repeat.
- Calculate the user's age by subtracting the birth year from the current year.
- Loop until a valid residence status (R or N) is entered:
 - Prompt the user for their residence status.
 - Read the residence status and convert it to uppercase.
 - Validate the residence status input.
- Loop until a valid income value is entered:
 - Prompt the user for their chargeable income.
 - Read the income value.

- If the input is invalid (not a number) or less than the lowest income bound, display an error message and repeat.

3.2 Tax Calculation:

- If the user is a resident (residence status is 'R'):
 - Use a series of conditional statements to determine the tax bracket based on the income and apply the corresponding tax rate or flat rate (RFR1 to RFR8) to calculate the tax.
- If the user is a non-resident (residence status is 'N'):
 - Use a series of conditional statements like residents but with different flat rates (NFR1 to NFR8).

3.3 Display Results:

- Display a formatted output including the user's name with initial and last name, Employee ID, age, residence status, income, and calculated tax.
- Print a best regards message.

3.4 Recalculate Option:

- Prompt the user to enter 'Q' to quit or any key to re-calculate.
- Read the user input.

4. End Loop:

- Continue the do-while loop if the user input is not 'Q'.

5. Exit.

Exercise 2

1. Variables needed: fname (string), lname (string), gender (char), EmployeeID (string), birth_yr (int), age (int), residence (char), income (double), tax (double), UserInput (char), initial (char).
2. Variables used for input: fname, lname, gender, EmployeeID, birth_yr, residence, income.
3. Variable types:
 - fname and lname: string (to store the user's first and last name as text).
 - Employeeid : string (to user user's Employee ID as alphanumeric text).

- gender, residence: char (to store single characters representing gender, and residence status).
 - UserInput, initial: char (to store single characters representing the user's input and the initial of their first name).
 - birth_yr, age: int (to store the user's birth year and calculated age as whole numbers).
 - income, tax: double (to store decimal numbers representing the user's income and calculated tax amount).
4. The best loop to validate the input is a do-while loop. The loop condition can be based on the user's choice to re-calculate or quit (e.g., UserInput != 'Q'). This loop structure ensures that the program will execute at least once and allows the user to repeat the calculation if desired.

Exercise 3

➤ **Test case 1:**

Input:

First name: Anthony
Last name: Gordon
Gender: M
Employee ID: A09
Residential Status: N
Year Of Birth: 1997
Chargeable Income: \$82042.40

Output:

Dear Mr. A. Gordon (Employee ID: A09)

You are 27 years old and a Non-Resident of Fiji. Based on your provided income of \$82042.40, you are required to pay an Income Tax of \$16408.48.

Best Regards.

➤ **Test Case 2:**

Input:

First name: Siara
Last name: Gordon
Gender: F
Employee ID: N182
Residential Status: R
Year Of Birth: 1999
Chargeable Income: \$82042.40

Output:

Dear Ms. S. Gordon (Employee ID: N182)

You are 25 years old and a Resident of Fiji. Based on your provided income of \$82042.40, you are required to pay an Income Tax of \$10008.48.

Best Regards.

➤ **Test Case 3:**

Input:

First name: Alexender
Last name: Reyes
Gender: O
Employee ID: P4136
Residential Status: R
Year Of Birth: 2002
Chargeable Income: \$30001

Output:

Dear Mx. A. Reyes (Employee ID: P4136)

You are 22 years old and a Resident of Fiji. Based on your provided income of \$30001, you are required to pay an Income Tax of \$0.18.

Best Regards.

Test Cases 1 and 2 were not satisfied since the calculated value calculated value was rounded off to nearest tenths (i.e. First decimal place), in the program output. However, this

didn't happen with **Test Case 3**. This was satisfied since the calculated value was equal to the value of the output of the program.

Exercise 4

Making a biased report in the program, such as reporting more income or less tax, would not be ethical and would violate both the ACS Code of Ethics and basic Programming Ethics.

Ethics in programming requires programmers to adhere to principles of honesty, integrity, and transparency. The ACS Code of Ethics, which sets the standards for professional conduct in the field of ICT (Information and Communication Technology), emphasizes the importance of acting honestly and with integrity. It specifically states that ICT professionals should be honest in their representation of skills, knowledge, services and products. By intentionally distorting the reported income or tax information, the program would be providing false and misleading information, which goes against the principles of honesty and integrity.

In summary, it is not ethical to make a biased report in the program by manipulating the reported income or tax information. Such actions would contradict the principles of honesty, integrity, and accuracy that are expected in the field of programming and would violate both the ACS Code of Ethics and basic Programming Ethics.

Exercise 5

1. (a) I did not copy from anyone's work. However, I did some research on loops on the concepts which were difficult, to have a fair idea on it.
(b) I never allowed anyone to copy from mine, but I did help a few of my mates to help understand about how some loops work loops.
2. - Yes, I did place my name and student number on the first page.
- Yes, I did place my name and student number in the program.
- Yes, I will upload a write-up for Sections 2 & 4.
- Yes, I did upload the Program for Section 3.
- Yes, I did submit before the deadline.

